

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5003		
Subject Title	Integrated Circuits Design		
Credit Value	3		
Level	5		
Pre-requisite / Co-requisite/ Exclusion	Nil		
Objectives	This subject aims to train students with the fundamental knowledge of (digital and analog) integrated circuit design in the advanced CMOS process. The student will get hands-on experience to design and analyze analog CMOS integrated circuits. The complete analog IC design flow including schematic design, simulation, layout design, and parasitic extraction using EDA tools in the industry will be covered in the course. The basic digital IC design concept and System-on-Chip (SoC) design flow including circuit design and verification, design for test (DFT), and physical design will also be introduced.		
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) basic knowledge of digital integrated circuit and SoC design flow in advanced CMOS process. (b) define the basic building blocks of analog integrated circuits, including single-gain-stage differential amplifiers, multi-stage amplifier, bandgap reference, etc.; (c) analyze and compute the behaviors of various amplifiers including voltage gain, frequency response, unity-gain bandwidth, and power consumption; (d) make design trade-off in analog circuit design for noise, mismatch, nonlinearity, stability, etc. in advanced CMOS process; (e) use mainstream EDA tools in the industry to optimize the performance of analog module circuits; and		
Subject Synopsis/ Indicative Syllabus	Digital and SoC design flow: Design flow for digital circuit and SoC in advanced CMOS process in the industry. Current mirror and basic amplifier: basic current mirror, cascode current mirror, gain-boost current mirror, common-source amplifier, and source follower. Voltage reference: low voltage bandgap voltage reference, current reference and temperature sensor. Advanced amplifiers: multi-stage amplifier, folded-cascode amplifier, differential amplifier with common-mode feedback, class AB amplifier and their applications.		
Teaching/Learning Methodology	Lecture: The design and characteristics of different analog integrated circuits will be explained. Concrete examples will be given to illustrate the difference between the different designs of advanced CMOS devices. Assignment sets will be given to assess the learning progress of students. Computer Laboratory: Students will work on problem sets and circuit/layout design excises in the tutorials, which provide them opportunities to apply the knowledge gained in lectures with EDA tools for IC design in the industry.		
Assessment Methods in	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)

Alignment with Intended Learning Outcomes			a	b	c	d	e
	1. Continuous Assessments (Assignments and tests)	50%	✓	✓	✓	✓	✓
	2. Project	50%		✓	✓	✓	✓
	Total	100 %					
Continuous assessments, which include assignments and test, are aimed at checking the progress of students throughout the course and assisting the students in self-monitoring their performances. A project, which is related to the design and analysis of integrated circuits, will be assigned to a group of students during the second half of the semester. The students will use their knowledge gained in the class/lab to design an analog module and to tackle a problem related to the topic of studies.							
Student Study Effort Expected	Class contact:						
	▪ Lecture					26 Hrs.	
	▪ Computer Lab (Project)					13 Hrs	
	Other student study efforts:						
	▪ Reading and self-study					81 Hrs.	
	The total student study effort					120 Hrs.	
Reading List and References	Jan M. Rabaey, et al. Digital integrated circuits: a design perspective (2nd Edition), Pearson Education, 2003 (https://www.pearson.com/uk/educators/higher-education-educators/program/Rabaey-Digital-Integrated-Circuits-2nd-Edition/PGM476523.html) Willy M. Sansen, Analog design essentials. Springer, 2006. Paul R. Gray, et al. Analysis and design of analog integrated circuits (Fifth Edition). John Wiley & Sons, 2009. Behzad Razavi, Design of Analog CMOS integrated circuits (Second Edition). McGraw-Hill Education, 2017.						